

A Microfluidic Hydrogel-Based Platform for Modeling Transport-Limited Cellular Environments in Diabetes

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Abstract

Diabetic microvascular complications are strongly influenced by the transport of glucose and signaling molecules across vascular barriers, yet many conventional in vitro disease models expose cells to static bulk concentrations that do not reproduce physiologically relevant flow and diffusion behavior. To address this limitation, a modular microfluidic platform was developed to decouple vascular perfusion from tissue exposure using a perfused serpentine channel separated from downstream culture chambers by a tunable hydrogel transport barrier. The device was fabricated by combining 3D-printed molds with PDMS soft lithography, and passive hydrostatic pressure was used to generate continuous flow through the vascular channel. Computational fluid dynamics simulations were used to characterize pressure, velocity, and wall shear stress profiles, while grayscale-controlled photopolymerization was used to fabricate hydrogel barriers with adjustable transport properties. Experimental characterization demonstrated stable hydrostatic perfusion at approximately 10 $\mu\text{L}/\text{min}$ with laminar flow conditions. Hydrogel thickness after equilibration depended on grayscale exposure, with mean final thicknesses of 0.768 mm, 0.656 mm, and 0.511 mm for grayscale values of 255, 200, and 150, respectively. Glucose diffusion studies showed that lower grayscale hydrogels produced greater solute transport, with normalized relative effective diffusivity values of 1.00, 1.23, and 1.70 across the same conditions. These results demonstrate that grayscale photopolymerization can be used to tune transport behavior within microfluidic systems and that the developed platform provides a practical framework for constructing more physiologically realistic vascularized disease models for diabetes research, drug screening, and future endothelialized organ-on-chip studies.

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Contribution Statement

This thesis was completed alongside Stone Fisher, who led CFD modeling and aided in CAD design, chip and hydrogel fabrication, and manuscript preparation. Josh Levin led conceptual development, CAD design, fabrication, experimental testing, data analysis,

and manuscript preparation under faculty supervision.

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Chapter 1

Introduction

Diabetes mellitus is a systemic metabolic disorder characterized by chronic hyperglycemia and widespread vascular and metabolic dysfunction. While its complications in the cardiovascular, renal, and nervous systems are well characterized, the effects of diabetes on skeletal health remain incompletely understood (Hofbauer et al. 2022; Napoli et al. 2017). Clinically, both type 1 and type 2 diabetes are associated with increased fracture risk and impaired bone healing; paradoxically, individuals with type 2 diabetes often present with normal or elevated bone mineral density despite heightened fragility (Vestergaard 2007; Schwartz 2016). This discrepancy suggests that diabetes impairs bone quality rather than bone quantity alone. However, the mechanisms underlying diabetes-associated skeletal degeneration remain unresolved.

Bone is a metabolically active and highly vascularized tissue that relies on continuous nutrient supply to sustain osteoblast, osteoclast, and osteocyte function. Glucose serves as a primary metabolic substrate for bone cells and plays an essential role in osteoblast differentiation and matrix production (Janghorbani et al. 2007; Karner and Long 2017). In diabetes, systemic glucose dysregulation leads to persistent hyperglycemia, increased formation of advanced glycation end products (AGEs), oxidative stress, and endothelial dysfunction, all of which have been implicated in altered bone remodeling (Yamamoto and Sugimoto 2016; Jain 1987). Yet it remains unclear whether skeletal dysfunction arises

primarily from elevated glucose concentration itself, from altered vascular transport, from AGE-mediated matrix stiffening, or from a combination of these factors.

A critical but often overlooked variable in experimental models is the mechanism of glucose delivery. In vivo, glucose is transported through the vascular network and diffuses across capillary walls before reaching bone cells. This transport process is governed by vascular flow, endothelial permeability, and diffusion gradients (Michel and Curry 1999; Prisby 2017). In contrast, many in vitro studies model hyperglycemia by directly supplementing cell culture media with high glucose concentrations, effectively bypassing the vascular transport step. While experimentally convenient, this approach assumes that concentration alone determines cellular response and neglects the role of diffusion-mediated transport dynamics. Whether the mode of glucose delivery influences cellular behavior in diabetic bone disease remains largely unexplored.

Direct investigation of localized glucose diffusion in vivo is technically challenging due to bone’s dense mineralized matrix and complex vascular architecture (Prisby 2017). Animal models provide systemic insight but do not easily allow isolation of microvascular transport variables, and ethical and cost considerations limit their scalability. Traditional two-dimensional cell culture models lack a vascular interface altogether and therefore fail to replicate physiologically relevant nutrient transport. Although bone-on-a-chip platforms have emerged to model bone physiology and drug responses (Y. Zhang et al. 2023; B. Zhang et al. 2018), few systems are specifically designed to decouple convective flow from diffusion through a tunable vascular barrier under diabetic conditions. As a result, a mechanistically focused platform for studying vascular-mediated glucose diffusion in bone is needed.

In physiological conditions, glucose transport from blood to tissue occurs via convective flow within vessels followed by diffusion across the capillary endothelium and basement membrane. Accurate modeling of this process requires spatial separation of flow and diffusion through a controllable barrier that approximates capillary permeability. Many existing microfluidic systems do not incorporate a tunable diffusion barrier that mimics

capillary basement membrane properties, limiting their ability to investigate transport-mediated cellular responses. Addressing this gap is essential for determining whether diffusion dynamics contribute to skeletal dysfunction in diabetes.

The objective of this work is to design and validate a microfluidic bone-on-a-chip platform capable of modeling glucose diffusion through a hydrogel-based barrier into cell compartments under physiologically relevant conditions. By focusing specifically on vascular-mediated glucose transport rather than full organ complexity, this system aims to isolate a potentially critical but understudied mechanism in diabetic skeletal pathology.

The proposed device incorporates a tunable hydrogel diffusion barrier positioned between a perfused vascular channel and cell culture compartments. Glucose-rich media flows through the vascular channel under laminar conditions, while solute transport into the cell chambers occurs exclusively via diffusion across the hydrogel barrier. This configuration separates convective transport from cellular exposure, thereby more closely approximating in vivo nutrient delivery. The hydrogel barrier is engineered to be tunable to relevant permeability, enabling controlled investigation of concentration gradients and diffusion kinetics.

This study employs both experimental and computational approaches. A physical microfluidic device is fabricated and characterized, while computational fluid dynamics (CFD) simulations are used to model flow profiles, shear stress, and pressure gradients. CFD-guided design optimization ensures laminar flow and controlled diffusion prior to fabrication. The integration of experimental validation and computational modeling enables systematic analysis of glucose transport mechanisms within a controllable microenvironment.

While this work is motivated by diabetic skeletal dysfunction, the platform is not limited to studying disease mechanisms alone. By reconstructing capillary-mediated transport in a controlled microfluidic system, this approach enables evaluation of how biological systems and potential therapies behave under physiologically realistic delivery conditions.

In particular, therapies that perform well under uniform in vitro exposure may exhibit

altered efficacy when subject to diffusion-limited transport, spatial gradients, and temporal delays. Capturing these effects is essential for improving the predictive accuracy of in vitro models.

Thus, beyond its mechanistic focus, this system establishes a foundation for transport-informed drug testing and personalized disease modeling, where both biological response and delivery dynamics can be systematically controlled and evaluated.

Chapter 2

Theoretical Background

2.1 Literature Review

2.1.1 Organ-on-a-Chip (OOC) Technology

Current drug discovery and toxicological research are limited by a lack of reliable predictive models that can accurately assess human physiological responses (Van Norman 2019). Animal models remain a primary source for pharmacokinetic data because they capture complex *in vivo* physiology, but they often fail to predict human outcomes due to species-specific metabolic and physiological differences, in addition to raising significant ethical concerns (Seok, Warren, Cuenca, et al. 2013). Conversely, traditional 2D *in vitro* models are simple and low-cost but fail to replicate the dynamic 3D cellular microenvironment, resulting in cells that lack the morphology, polarity, and behavior observed *in vivo* (Edmondson et al. 2014).

Organ-on-a-Chip (OOC) technology has emerged as a bridge between these methods, combining micro-engineering with 3D cell culture to recapitulate organ-level functions. By utilizing human cells in dynamic microfluidic environments, these devices provide the mechanical and chemical cues (such as fluid flow and shear stress) necessary for more accurate personalized disease modeling and drug screening. Bone-on-a-chip platforms specifically allow for the investigation of bone adaptation, repair, and diseases like osteoporosis within

a controlled, biomimetic system.

2.1.2 The Basement Membrane and Capillary Interface

The basement membrane (BM) is a specialized extracellular matrix structure that underlies endothelial and epithelial tissues, forming a selective barrier between the vascular lumen and surrounding parenchyma (Yurchenco 2011). Ultrastructurally, it consists of the lamina lucida, lamina densa, and lamina fibroreticularis. The lamina densa, primarily composed of type IV collagen and laminins, provides mechanical support and contributes significantly to molecular filtration properties (Yurchenco 2011).

In bone, nutrient exchange occurs through a highly vascularized network that includes sinusoidal capillaries within the marrow space (Williamson, Kilo, and Tilton 1993). Capillary-mediated transport involves convective flow within vessels followed by diffusion across the endothelial layer and basement membrane before solutes reach interstitial tissue. Transport is governed by permeability, surface area, and concentration gradients (Bird, Stewart, and Lightfoot 2002).

In diabetic conditions, microvascular dysfunction and basement membrane thickening have been documented across multiple tissues (Brownlee 2001). These structural alterations may influence solute flux into tissues; however, most in vitro diabetic bone models bypass the vascular interface entirely by directly exposing cells to hyperglycemic media. Consequently, the influence of capillary-mediated diffusion dynamics on skeletal cell behavior remains poorly characterized

2.1.3 Hydrogels as Biomimetic Scaffolds

Hydrogels are three-dimensional crosslinked polymer networks capable of retaining large volumes of water, making them structurally and mechanically similar to native ECM (Peppas et al. 2000). Their high water content supports diffusion of small solutes, while their mesh size and crosslinking density regulate permeability and mechanical stiffness.

Photocrosslinkable materials such as polyethylene glycol diacrylate (PEGDA) and

gelatin methacryloyl (GelMA) are widely used in microfluidic and tissue-engineered systems due to their tunable mechanical properties, cytocompatibility, and optical transparency (Zhu and Marchant 2011; Yue et al. 2015). PEG-based hydrogels provide controlled diffusivity with minimal protein adsorption, whereas GelMA incorporates bioactive motifs that promote cell adhesion.

In this work, a hydrogel layer functions as an artificial basement membrane positioned between a perfused vascular channel and cell compartments. By tuning polymer concentration and crosslinking density, the hydrogel permeability can be adjusted to approximate physiological capillary transport while maintaining mechanical stability. Unlike rigid membranes, hydrogel barriers more closely resemble the compliant and hydrated structure of native capillary interfaces.

2.2 Governing Physics

2.2.1 Mass Transport and Diffusion Kinetics

Solute transport across the hydrogel barrier is governed primarily by passive diffusion. For time-dependent systems, concentration profiles are described by Fick’s Second Law:

$$\frac{\partial C}{\partial t} = D \nabla^2 C$$

where C is solute concentration, D is the effective diffusion coefficient within the hydrogel, and t represents time.

Within polymeric networks, effective diffusion coefficients are reduced relative to free solution and depend on mesh size, tortuosity, and solute–polymer interactions (Bird, Stewart, and Lightfoot 2002). Small molecules such as glucose ($\approx 180Da$) typically experience modest hindrance, whereas larger macromolecules may exhibit restricted transport.

Transport selectivity in hydrogels arises through two primary factors: size exclusion determined by the relationship between solute hydrodynamic radius and mesh size, and interaction-based filtering, including electrostatic and hydrophobic interactions that alter effective diffusivity independent of size (Bird, Stewart, and Lightfoot 2002; Amsden 1998).

When modeling small diffusion cells, lateral diffusion at the membrane edges (“edge effects”) can artificially elevate apparent steady-state flux if not properly constrained, particularly when the effective diffusion area exceeds the exposed cross-sectional region (Bird, Stewart, and Lightfoot 2002). Device geometry must therefore be incorporated into computational modeling.

2.2.2 Computational Fluid Dynamics and Laminar Flow

Microfluidic systems typically operate at low Reynolds numbers ($Re < 1-100$), where viscous forces dominate inertial forces and flow remains laminar (Whitesides 2006). Under laminar conditions, velocity profiles are predictable and mixing occurs primarily by diffusion rather than turbulence.

CFD simulations are employed to validate laminar flow conditions within the vascular channel, predict wall shear stress experienced by endothelial cells, model velocity and pressure gradients, and optimize channel geometry to ensure uniform concentration along the barrier.

For perfused systems with sufficiently high flow rates relative to diffusion flux, the donor channel may be approximated as an infinite source, maintaining a near-constant concentration at the hydrogel boundary (infinite dose assumption).

2.2.3 Glucose Transport and Utilization Modeling

Glucose transport within biological systems involves both passive diffusion through extracellular environments and active cellular uptake. In many cell types, glucose uptake is mediated by facilitative transporters such as GLUT proteins and can be described using Michaelis–Menten kinetics, where uptake rate depends on local glucose concentration. Glucose uptake in bone cells is mediated primarily by facilitative glucose transporters such as GLUT1 and, under certain conditions, GLUT4 (Brownlee 2001; Williamson, Kilo, and Tilton 1993).

In systems where transport and consumption are coupled, spatial concentration profiles

may be influenced not only by diffusion but also by cellular metabolic demand, potentially leading to localized depletion gradients. Modeling such behavior typically requires coupling diffusion equations with reaction terms representing cellular uptake.

In the present work, transport analysis focuses on diffusion-driven delivery across the hydrogel barrier, and cellular consumption is not explicitly modeled. This simplification allows isolation of transport dynamics as a primary variable. However, incorporation of coupled diffusion–consumption models represents an important extension for future studies, particularly in the context of evaluating metabolically active systems such as stem cell spheroids.

2.3 Theoretical Assumptions

To enable tractable analysis and computational modeling, several simplifying assumptions are applied to the system. These assumptions are selected to preserve the dominant physical behavior while allowing analytical and numerical treatment of transport processes.

The hydrogel barrier is not assumed to be globally homogeneous. In the present design, spatial variations in crosslinking density may be introduced through grayscale-controlled photopolymerization to approximate structural heterogeneity observed in native basement membranes. However, for modeling purposes, the hydrogel is treated as locally homogeneous, such that within sufficiently small regions the effective diffusion coefficient may be considered constant. This allows application of Fickian diffusion within each local domain while permitting spatial variation in transport properties at larger scales.

The effective diffusion coefficient D is therefore treated as a spatially varying parameter, $D=D(x,y,z)$, reflecting heterogeneity in mesh size and crosslinking density. In simplified models or initial approximations, an average or effective diffusion coefficient may be used to represent bulk transport behavior.

Solute transport across the hydrogel is assumed to occur primarily through passive diffusion, with negligible convective flow within the gel matrix. This assumption is justified by the low permeability and small pore size of the hydrogel, which suppresses bulk fluid

motion.

At the interface between the vascular channel and the hydrogel, solute concentration is assumed to be continuous, with no preferential partitioning or interfacial resistance. Additionally, due to continuous perfusion of the vascular channel, concentration at the hydrogel boundary is approximated as constant over time, consistent with an infinite source condition.

The hydrogel–cell chamber interface is assumed to allow unrestricted diffusion into the receptor compartment, without additional transport barriers. Cellular uptake and consumption of solutes are not included in the present transport model, allowing isolation of diffusion-driven behavior.

Finally, the system is assumed to operate under steady-state or quasi-steady-state conditions for flow, with transient effects considered only in diffusion analysis. These assumptions collectively enable focused investigation of transport dynamics while maintaining relevance to the physical behavior of the system.

Chapter 3

Methodology

3.1 System Overview

The development of the bone-on-a-chip microfluidic platform follows an iterative engineering design framework consisting of three sequential phases: (1) fabrication optimization, (2) transport and fluidic validation, and (3) biological characterization. The system is engineered to isolate and model capillary-mediated passive diffusion of soluble factors into a compartmentalized tissue microenvironment. While diabetic conditions serve as a primary application, the platform is designed more broadly as a transport-controlled testbed for evaluating biological response under physiologically relevant delivery constraints.

As shown in Figure 3.1, the device consists of a serpentine vascular channel (top layer) connected to inlet and outlet ports, positioned above multiple cell culture chambers (bottom layer). A hydrogel diffusion barrier is embedded at the interface between the channel and each chamber, permitting passive transport of soluble factors while preventing direct convective flow into the cellular compartment. This architecture enables controlled delivery of nutrients and signaling molecules, allowing cells to experience spatially and temporally varying concentrations that more closely mimic *in vivo* capillary-mediated transport.

Unlike conventional static 2D cultures, this system spatially separates convective flow from cellular compartments using a hydrogel-based diffusion barrier. By incorporating con-

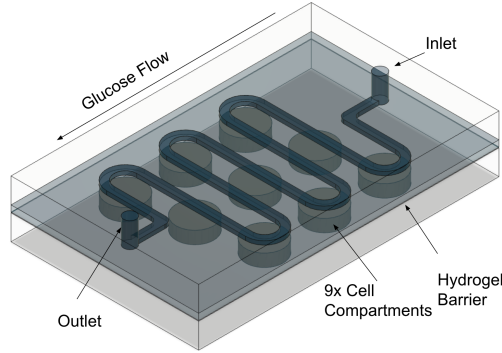


Figure 3.1: Schematic of the microfluidic bone-on-a-chip platform illustrating separation of convective flow and diffusive transport.

trolled laminar perfusion and tunable solute permeability, the platform enables mechanistic interrogation of vascular-mediated transport under diabetic conditions. This approach bridges the gap between reductionist *in vitro* systems and complex *in vivo* physiology while maintaining experimental controllability.

3.2 Device Design and Fabrication

3.2.1 CAD and Microfluidic Architecture

The final microfluidic platform developed in this study was preceded by several iterative design stages aimed at balancing physiological relevance, manufacturability, residence time, and experimental throughput. Because the primary objective of the system was to model vascular transport to downstream tissue chambers, the geometry of the perfused channel required progressive refinement to improve exposure dynamics while maintaining straightforward fabrication through soft lithography. The major design stages are summarized in Figure 3.2.

The first design iteration employed a simple linear perfusion channel positioned above two downstream cell compartments (Figure 3.2-A). In this concept, fluid entered through a single inlet, traveled through a straight vascular channel, and exited through a downstream outlet. Two circular culture chambers were aligned beneath the channel to receive diffusive transport through the hydrogel barrier separating the vascular and tissue regions.

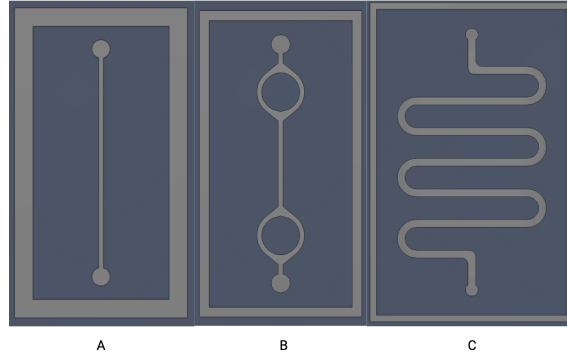


Figure 3.2: CAD designs of chip molds. (A) Initial linear perfusion design. (B) Intermediate dual-circle design. (C) Final serpentine design.

This design was selected as an initial prototype because of its simplicity and low fabrication risk. Straight channels are easy to model computationally, easy to mold using additive manufacturing, and less susceptible to flow irregularities or trapped air bubbles during early testing. Additionally, the minimal geometry allowed rapid evaluation of bonding methods, leakage behavior, hydrogel placement, and general perfusion feasibility before committing to more complex layouts.

However, the straight-channel design exhibited several conceptual limitations. Most notably, the short flow path produced limited fluid residence time above the downstream compartments, reducing the duration of local nutrient and solute exposure. The design also poorly represented the tortuous nature of physiological microvasculature and offered only two tissue compartments, restricting throughput and comparative studies. Because exposure conditions were likely dominated by rapid axial transport rather than prolonged exchange, a more developed geometry was pursued.

To increase path length while preserving a compact footprint, a second design iteration was developed in which the vascular channel looped around two circular regions before continuing downstream (Figure 3.2-B). In this architecture, the channel traced circular paths around each compartment, increasing total channel length relative to the straight design while maintaining two primary culture sites.

The rationale for this geometry was to enhance fluid residence time and create longer vascular contact regions surrounding each chamber. By increasing the time perfusate

spent adjacent to the tissue compartments, the design was expected to improve diffusive transport opportunities across the hydrogel barrier.

Despite these advantages, several limitations remained. The system still contained only two culture chambers, limiting experimental throughput. Channel length increased only modestly relative to later designs, and the circular loops did not fully utilize the available chip footprint. Additionally, the compartment arrangement was less modular and did not provide multiple spatial positions for comparing upstream versus downstream exposure effects. As a result, a higher-density architecture was pursued.

The final design iteration employed a serpentine vascular channel spanning the full device footprint, coupled with an expanded downstream chamber array (Figure 3.2-C). This geometry substantially increased channel length while maintaining a compact planar footprint and introduced repeated turns resembling the tortuous routing commonly observed in vascular networks.

The serpentine channel was selected for several reasons. First, the extended flow path increased hydraulic residence time, allowing solutes more time to diffuse across the hydrogel barrier before outlet removal. Second, the repeated curvature generated localized variations in velocity and wall shear stress that more closely resemble nonuniform physiological flow conditions than a straight channel. Third, the design used chip area efficiently, enabling integration of multiple tissue compartments without increasing device size.

The chamber array was expanded from earlier two-chamber designs to a total of nine compartments to improve throughput and enable spatial comparisons. Six chambers were positioned adjacent to curved serpentine segments, while three were located near relatively straight sections of channel. This arrangement was intentionally selected to investigate whether local residence time or geometric flow differences influenced downstream transport behavior. Curved segments were expected to exhibit slightly longer local interaction times and altered flow distributions compared with straightaways, potentially affecting solute delivery to neighboring chambers.

Although the serpentine design introduced greater fabrication complexity than the

earlier linear geometries, the increased channel length, higher compartment density, and improved physiological resemblance justified the transition. The final architecture provided the most versatile platform for studying transport-controlled cellular exposure while retaining compatibility with PDMS molding and hydrostatic pressure-driven perfusion.

Overall, the CAD development process demonstrates that the final chip geometry was not selected arbitrarily, but emerged through iterative engineering refinement in response to the competing requirements of transport performance, manufacturability, and experimental utility.

The microfluidic device is designed using Computer-Aided Design (CAD) software (Autodesk Fusion 360), incorporating established organ-on-a-chip architectural principles (Waheed et al. 2016). The platform consists of a sandwich design, with a serpentine perfused vascular channel (donor compartment) making up the top PDMS slab, a hydrogel barrier in the middle, and nine independent cell culture chambers (receptor compartments), arranged symmetrically with three chambers per side and three in the middle.

The vascular channel connects inlet and outlet ports to enable continuous perfusion of glucose-containing media under controlled flow rates. Each cell chamber interfaces with the vascular channel via a dedicated aperture engineered to house a hydrogel diffusion barrier. This barrier prevents convective mixing while permitting passive solute transport.

Channel geometry is designed to maintain laminar flow ($Re < 100$) and uniform solute concentration along the hydrogel interface. The cylindrical layout reduces concentration gradients between the upstream and downstream compartments.

3.2.2 3D Printing and Material Selection

Device fabrication is performed using Digital Light Processing (DLP) 3D printing due to its high resolution (10–30 μm), rapid prototyping capability, and reproducibility (Waheed et al. 2016)(Huh et al. 2010). Candidate materials include biocompatible photocurable resins and PDMS validated for microfluidic applications (Waheed et al. 2016).

Molding with Sylgard 184 PDMS (10:1 base:curing agent ratio) is used for fabrication.

PDMS remains widely used in microfluidics due to optical transparency, gas permeability, and ease of plasma bonding (Bhatia and Ingber 2014). Dimensional fidelity and channel integrity are verified by microcaliper measurement.

3.2.3 Hydrogel Barrier Fabrication and Integration

A semi-permeable hydrogel membrane functions as a biomimetic surrogate of the capillary basement membrane. Photocrosslinkable hydrogels such as PEGDA and GelMA are selected due to their tunable crosslink density, biocompatibility, and well-characterized diffusion properties.

Hydrogel precursor solution containing photoinitiator (LAP) is introduced into the DLP print vat. Controlled UV or visible-light exposure induces crosslinking, forming a stable membrane that is then placed between the PDMS slabs.

Hydrogel thickness, polymer concentration, and crosslinking time are systematically varied to modulate effective diffusion coefficients. Membrane formation is validated visually and through leakage testing prior to transport experiments.

3.3 Transport Validation and Data Acquisition

3.3.1 Computational Fluid Dynamics (CFD)

CFD simulations are conducted using OpenFOAM CFD to evaluate hydrodynamic behavior and solute transport. The Navier–Stokes equations are solved under steady-state laminar flow conditions (Amsden 1998).

Medium properties are modeled as water at 37°C, with dynamic viscosity (μ) = 7.75 $\times 10^{-4}$ kg/(m·s) and density (ρ) = 1000 kg/m³. Simulations are performed to confirm Reynolds number within laminar regime, quantify wall shear stress on endothelial surfaces and evaluate velocity uniformity along the hydrogel interface.

3.3.2 Diffusion and Permeability Assays

Quantitative characterization of hydrogel transport behavior required the development of a practical diffusion testing method compatible with the mechanically soft grayscale-printed hydrogel samples used in this study. Because these hydrogels were intended to function as transport barriers within the microfluidic platform, an experimental system was needed that could isolate the material between donor and receiver compartments while maintaining structural integrity during multi-hour testing.

Initial Franz Cell-Inspired Diffusion Chamber Design

The initial approach employed custom 3D-printed diffusion cells (see Figure 3.3) inspired by traditional Franz diffusion chambers. These devices were fabricated from polyethylene terephthalate glycol (PETG) and consisted of donor and receiver reservoirs designed to sandwich the hydrogel sample between two rigid printed halves. Mechanical compression was applied using binder clips to seal the assembly and prevent leakage.

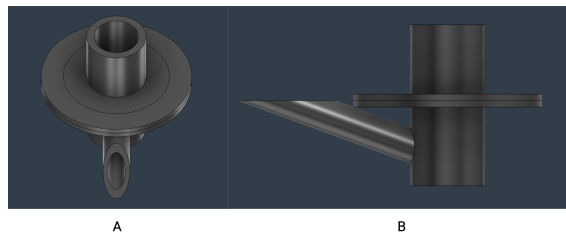


Figure 3.3: CAD design of 3D-printed Franz cell diffusion rig. (A) Front view showing sampling port and top donor compartment. (B) Side view showing assembly; the hydrogel is placed between the top and bottom compartments and mechanically clamped between them.

This method offered several theoretical advantages. First, it provided clear physical separation between donor and receiver chambers, enabling one-dimensional transport through the hydrogel barrier. Second, the clamped design was low-cost, reusable, and rapidly fabricated using standard additive manufacturing methods. Third, the chamber geometry allowed straightforward filling and sampling of both compartments.

However, repeated testing revealed a critical limitation: the grayscale-defined hydrogels

lacked sufficient mechanical robustness to withstand sustained compressive loading at the sealing interface. Even moderate binder clip pressure frequently caused tearing, cracking, edge deformation, or rupture of the hydrated gels during assembly. Because seal integrity depended directly on compression force, reducing clamp pressure often led to leakage, while increasing pressure damaged the hydrogel. This tradeoff made the Franz cell approach unreliable for fragile hydrogel formulations.

Following these failures, several additional mechanically clamped testing concepts were considered, including using syringes as donor and receptor chambers. While these designs could potentially lower local stress concentrations, they retained the same fundamental challenge: accurate diffusion measurements required leak-free compartment separation, while the hydrogels could not reliably tolerate compressive sealing loads. Given project time constraints and limited material availability, continued iteration on compression-based fixtures was deprioritized.

Final Six-Well Plate Drop Diffusion Method

A simplified static diffusion assay was therefore adopted using standard six-well culture plates. In this final configuration, individual hydrogels were placed in wells containing a defined volume of media as the receiver. A concentrated donor droplet containing glucose solution was then deposited directly on the upper hydrogel surface.

In this arrangement, glucose transport occurred from the donor droplet through the hydrogel thickness into the surrounding receiver media below. Receiver-phase glucose concentration was measured at scheduled time points using a glucometer to compare relative transport behavior across grayscale conditions.

The six-well plate method was selected because it eliminated the need for mechanical compression, substantially reducing hydrogel damage and sample loss. It also enabled parallel testing of multiple conditions and replicates using widely available laboratory materials. Sample loading, retrieval, and handling were significantly faster than with custom clamped fixtures, improving experimental throughput during late-stage thesis testing.

The final method was intentionally selected as a comparative transport assay rather than a perfect classical diffusion cell analog. Its major benefits included no compressive stress on fragile hydrogels, rapid parallel testing of multiple grayscale conditions, minimal custom hardware requirements, and reduced leakage risk relative to clamped fixtures. However, its limitations included less strictly defined one-dimensional transport geometry, possible donor droplet spreading variability, potential evaporative effects during long-duration tests, and greater sensitivity to hydrogel placement consistency. This made it more appropriate for relative diffusivity comparisons than absolute permeability extraction.

Despite these limitations, the method was sufficient to resolve reproducible grayscale-dependent transport trends and therefore served to determine whether grayscale-controlled photopolymerization could be used to tune hydrogel barrier transport properties.

For each condition, equilibrated hydrogels were transferred into six-well plates containing receiver media of known volume. A glucose donor droplet of fixed volume and concentration was applied to the hydrogel surface using the pipette guide fixture. Receiver media glucose concentration was then measured at predetermined intervals over 48 hours. Triplicate samples were tested for each grayscale condition (255, 200, and 150), and resulting concentration profiles were used to estimate relative effective diffusivity after correction for measured hydrogel thickness.

3.3.3 Hydrostatic Pressure-Driven Flow

Continuous flow within the microfluidic device is achieved using a hydrostatic pressure-driven setup. Fluid reservoirs connected to the inlet and outlet ports are positioned at different vertical heights to establish a gravitational pressure gradient across the device.

The inlet reservoir consists of a vertically mounted syringe or fluid column containing glucose- or solute-enriched media. The outlet tubing is directed to a waste reservoir positioned at a lower elevation. The resulting height difference between the fluid levels generates a pressure differential that drives flow through the vascular channel.

To ensure reproducibility across experiments, all devices are operated under identical height differences, and tubing lengths and diameters are kept consistent to maintain uniform hydraulic resistance. Prior to each experiment, all channels and tubing are primed with media to eliminate air bubbles, which can disrupt flow stability.

Flow rates are estimated by collecting effluent over a fixed time interval and calculating the volumetric flow rate. These measurements are used to verify that experimental flow conditions fall within the laminar regime predicted by CFD simulations.

To minimize temporal variation in flow rate due to depletion of the inlet reservoir, relatively large fluid volumes are used such that the change in hydrostatic head over the course of the experiment remains small. This approach enables sustained, low-Reynolds-number perfusion suitable for long-duration transport and cell culture studies.

Chapter 4

Results

4.1 Device Fabrication and Dimensional Validation

The microfluidic platform was successfully fabricated using a mold-based soft lithography workflow designed to produce a vascular serpentine channel adjacent to a larger downstream cell chamber separated by a hydrogel interface region. The device geometry was first designed in CAD software and exported as a stereolithography (STL) model for additive manufacturing of the mold. Following mold fabrication, polydimethylsiloxane (PDMS) was cast, cured, and removed from the mold to generate the final microfluidic layer. Inlet and outlet ports were created with syringe tips to permit fluidic connection to external reservoirs, and the device was subsequently bonded to a flat substrate to complete the enclosed channel architecture. Representative images of the design, printed mold, and cured PDMS chips are shown in Figure 4.1.

Measured device dimensions were compared to the nominal geometry defined in the CAD model to assess fabrication accuracy. The microchannel had sub-millimeter dimensions, with an average height of 0.468 ± 0.022 mm and width of 0.881 ± 0.105 mm, consistent with the design intent. The total channel length was measured to be approximately 21.01 mm, aligning with the expected serpentine geometry used to increase residence time. The cell compartment dimensions (3.038 ± 0.030 mm width and 1.354 ± 0.008 mm height) were also consistent across samples and matched the intended expansion of the diffusion

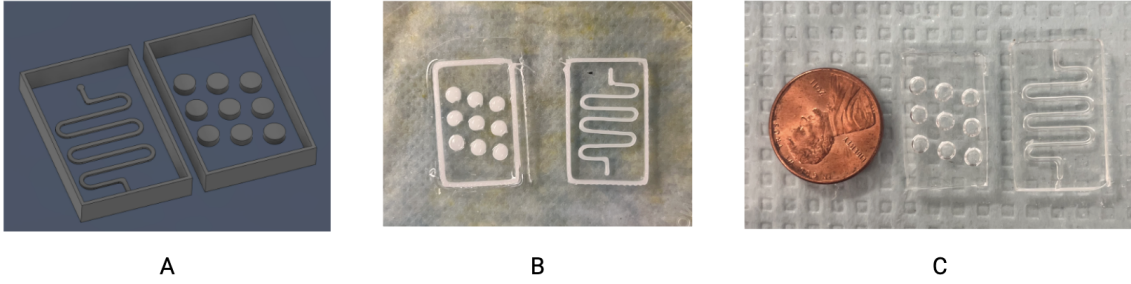


Figure 4.1: Fabrication workflow for the transport-controlled microfluidic platform. (A) Computer-aided design renderings of the mold geometries used to create the serpentine vascular channel and downstream chamber structures. (B) Fabricated molds produced by additive manufacturing. (C) Final transparent PDMS device components following casting and demolding, shown with a U.S. penny for size reference.

region relative to the microchannel. Minor deviations from the nominal design are attributed to fabrication limitations and measurement uncertainty at small scales, but these differences are not expected to impact flow or transport behavior significantly. Overall, the fabricated device dimensions demonstrate acceptable agreement with the design specifications and support the validity of subsequent experimental and computational analyses.

4.2 Flow Characterization and Validation of the Vascular Channel

Hydrodynamic performance of the vascular channel was evaluated to determine whether the fabricated device could sustain stable perfusion under physiologically relevant microfluidic conditions. Flow through the serpentine channel was driven passively using a hydrostatic pressure gradient established by a height difference between inlet and outlet reservoirs (See Figure 4.2). This approach eliminated the need for external pumps while enabling steady low-flow operation over extended time periods.

The measured average flow rate under standard operating conditions was approximately 10 $\mu\text{L}/\text{min}$ based on outlet volume collection. Given the measured channel cross-sectional dimensions ($0.881 \text{ mm width} \times 0.468 \text{ mm height}$), this corresponds to an average linear velocity of approximately 0.40 mm/s within the channel. This velocity lies within

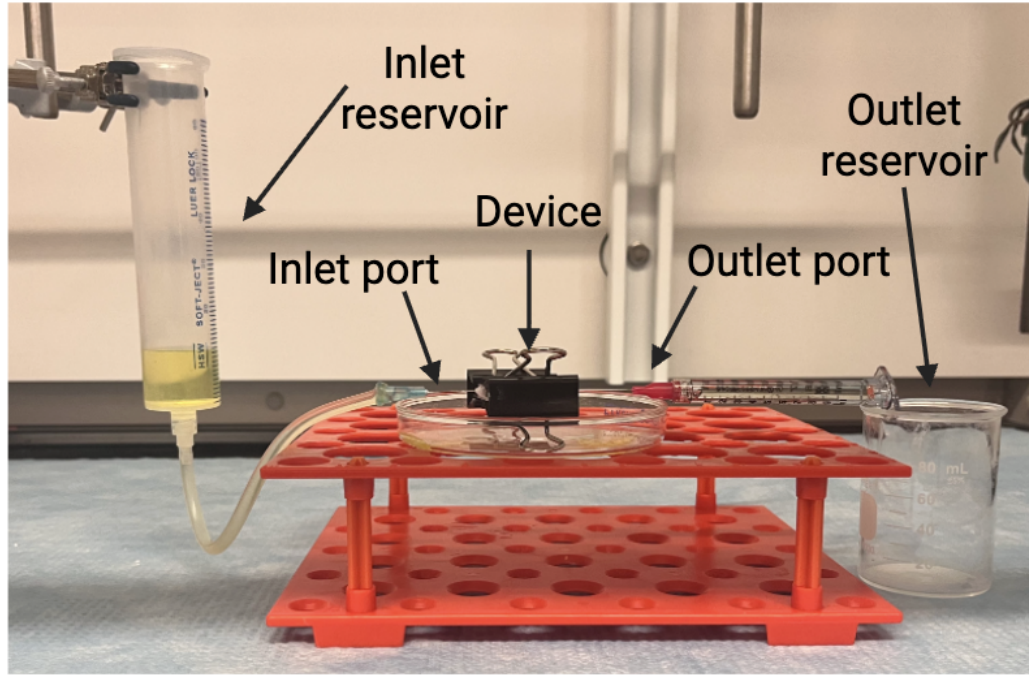


Figure 4.2: Hydrostatic perfusion setup used for passive flow generation through the microfluidic device. Side view of the experimental apparatus showing an elevated inlet reservoir connected to the device inlet and a downstream outlet collection vessel, generating flow through a hydrostatic pressure difference due to 1.0 inch height difference.

the lower range reported for physiological capillary microcirculation, where resting capillary blood velocities on the order of 0.3–1.8 mm/s have been observed, and is therefore consistent with low-shear microvascular transport environments (Hudetz 1997). As a result, the selected operating condition was considered suitable for mimicking transport conditions relevant to nutrient delivery in vivo.

The Reynolds number was estimated using measured geometry and water-like fluid properties:

$$Re = \frac{\rho U D}{\mu}$$

where U is the mean velocity, and D is the hydraulic diameter of the rectangular channel. Under the measured flow condition, the Reynolds number remained well below unity, confirming strongly laminar flow dominated by viscous forces. This behavior is

expected in microfluidic systems and is advantageous because it produces predictable transport conditions without turbulence or flow instabilities.

Computational fluid dynamics simulations further supported the experimental observations. Velocity contours demonstrated a relatively uniform axial flow profile through the straight channel segments, with localized increases in velocity near channel bends due to curvature-induced redistribution of streamlines. Pressure decreased monotonically along the length of the serpentine path, indicating consistent hydraulic resistance throughout the device. Streamline plots showed smooth trajectories without recirculation zones or stagnation regions.

The serpentine geometry increased the effective channel length to approximately 80 mm while maintaining a compact device footprint. This extended path length provided two functional advantages: increased hydraulic resistance, allowing physiologically relevant low flow rates to be achieved using modest hydrostatic pressure, and increased residence time, permitting prolonged interaction between the flowing media and the hydrogel transport interface.

At the measured average velocity of 0.40 mm/s, the estimated fluid residence time through the vascular channel was approximately:

$$t = \frac{L}{U} = \frac{80mm}{0.40mm/s} \approx 200s.$$

or roughly 3.3 minutes. This residence time is sufficient to support sustained solute exchange along the hydrogel interface during perfusion.

Overall, both experimental measurements and computational modeling indicate that the device successfully supports stable, low-Reynolds-number perfusion in a physiologically relevant flow regime. These results validate the serpentine vascular channel as an appropriate upstream transport module for subsequent diffusion-controlled studies.

Pressure distribution along the serpentine channel confirms a stable pressure-driven flow regime. As shown in Figure 4.3, pressure decreases monotonically from inlet to outlet, with no abrupt discontinuities or localized regions of stagnation. This smooth gradient indicates that flow is governed by viscous forces and that the system operates

under steady-state conditions. The absence of pressure irregularities ensures a consistent driving force for fluid motion, which is critical for maintaining uniform solute delivery along the hydrogel interface.

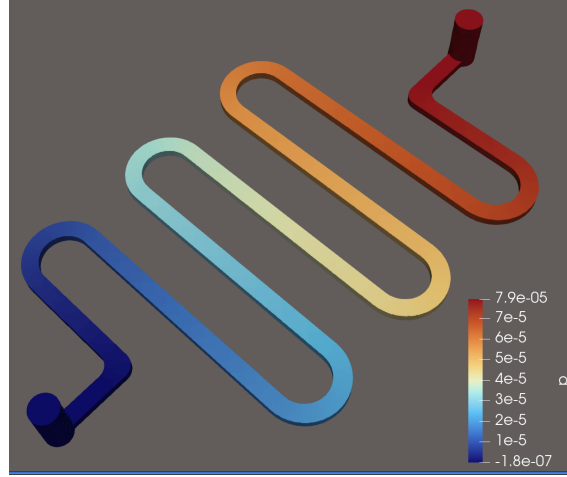


Figure 4.3: Simulated pressure distribution along the serpentine vascular channel under steady-state laminar flow conditions (Pa).

Flow structure within the channel further supports laminar behavior. Streamline visualization (Figure 4.4) demonstrates smooth, continuous flow paths that closely follow the channel geometry, including through curved segments. No recirculation zones or vortices are observed, indicating that inertial effects are negligible and that mixing within the channel occurs primarily through diffusion rather than turbulence. This is consistent with the low Reynolds number regime typical of microfluidic systems and ensures predictable transport of solutes to the diffusion barrier.

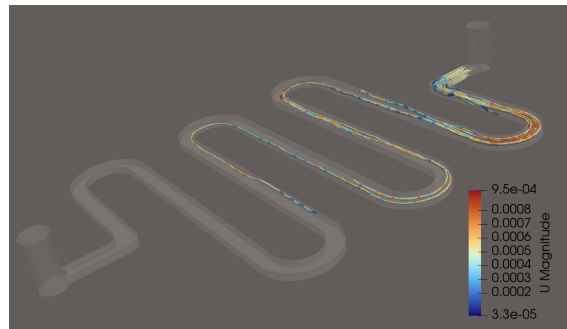


Figure 4.4: Streamline visualization of fluid flow within the vascular channel (m/s).

Velocity field analysis provides additional confirmation of stable flow conditions. As shown in Figure 4.5, velocity profiles exhibit characteristic laminar behavior, with maximum velocities at the channel center and reduced velocities near the walls due to viscous drag. Importantly, velocity remains relatively uniform along the length of the channel, despite the serpentine geometry. This spatial consistency supports the assumption of a near-constant solute concentration at the hydrogel boundary, justifying treatment of the vascular channel as an effectively well-mixed donor compartment in subsequent transport analysis.

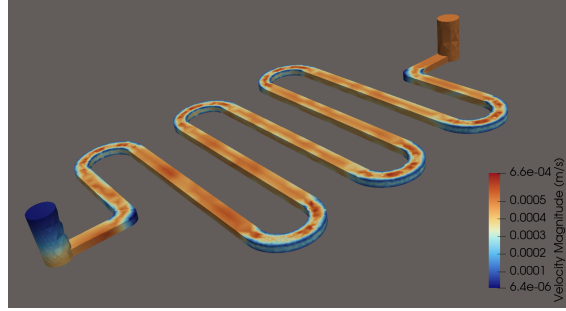


Figure 4.5: Velocity magnitude distribution within the vascular channel.

Wall shear stress was evaluated to characterize mechanical conditions at the channel interface. As shown in Figure 4.6, shear stress is elevated at channel bends, where directional changes in flow induce higher velocity gradients, and reduced in straight sections. Overall shear levels remain low and within ranges typical of microfluidic perfusion systems. While endothelial cells are not incorporated in the current study, these values suggest that the system is compatible with future integration of vascular cell layers without inducing excessive mechanical stress.

Taken together, these results demonstrate that the vascular channel provides stable, laminar, and spatially consistent flow conditions suitable for controlled solute delivery. The absence of recirculation, uniform velocity distribution, and predictable pressure gradient support the use of this system as a well-defined upstream boundary for diffusion-driven transport across the hydrogel barrier.

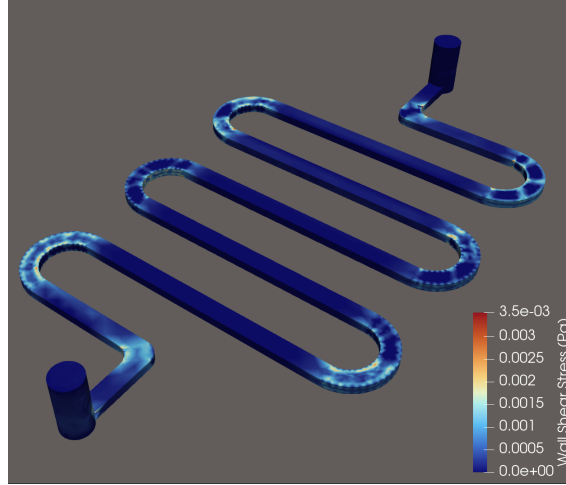


Figure 4.6: Wall shear stress distribution along the vascular channel.

4.3 Hydrogel Characterization and Diffusion Tuning

The hydrogel interface serves as the central transport-regulating component of the microfluidic platform. In the present system, grayscale-controlled photopolymerization was used to generate hydrogels with differing effective crosslink densities, allowing transport behavior to be tuned through fabrication parameters rather than changes in bulk chemistry. Three representative grayscale conditions (255, 200, and 150) were evaluated to determine how grayscale exposure influenced hydrogel dimensional stability and glucose transport.

4.3.1 Hydrogel Dimensional Equilibration in Media

Prior to diffusion testing, hydrogels were conditioned in culture media to assess dimensional stability under hydrated operating conditions. Thickness and mass were measured over time as the materials equilibrated. Across all conditions, hydrogels exhibited time-dependent changes in geometry during the initial exposure period, followed by stabilization at later time points. Representative thickness trends are shown in Figure 4.7.

Final equilibrated thicknesses differed systematically across grayscale conditions. The highest grayscale condition (255) retained the greatest final thickness, while progressively lower grayscale values resulted in thinner final structures. Mean final thicknesses are

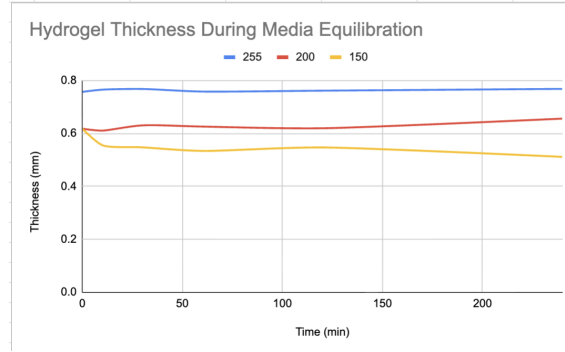


Figure 4.7: Mean hydrogel thickness over time during immersion in media for grayscale-defined constructs (255, 200, and 150), demonstrating condition-dependent dimensional stabilization.

Table 4.1: Final Mean Thicknesses of Grayscale Hydrogels

255	200	150
0.768 mm	0.656 mm	0.511 mm

shown in Table 4.1 and Figure 4.8.

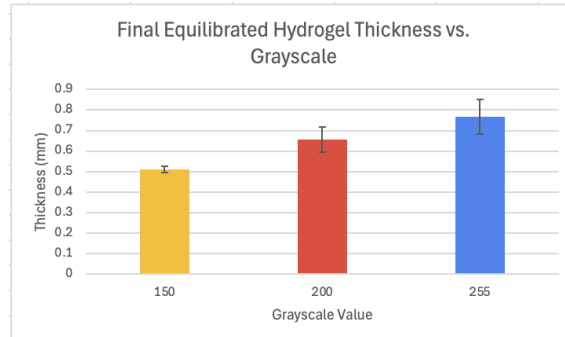


Figure 4.8: Mean final hydrated thickness after equilibration. Lower grayscale exposure produced thinner hydrogel barriers. Error bars represent standard deviation.

The observed decrease in final thickness with reduced grayscale exposure is consistent with lower photopolymerization dose and reduced network crosslink density. Less crosslinked hydrogels are expected to exhibit decreased structural rigidity and greater dimensional contraction or partial material loss during equilibration. In contrast, higher grayscale constructs retained greater structural integrity in media.

Mass measurements exhibited greater variability than thickness data, likely due to surface liquid retention and handling effects during weighing. As a result, thickness was

Table 4.2: Mean Receiver Concentrations at 10-hour t=Time Point

255	200	150
151 mg/dL	181 mg/dL	246 mg/dL

used as the primary geometric parameter for subsequent transport analysis.

4.3.2 Glucose Transport Across Grayscale-Defined Hydrogels

Glucose transport experiments were conducted by exposing one side of each hydrogel to glucose-containing media and monitoring concentration increases in the receiver phase over time. Mean glucose concentration profiles are shown in Figure 4.9.

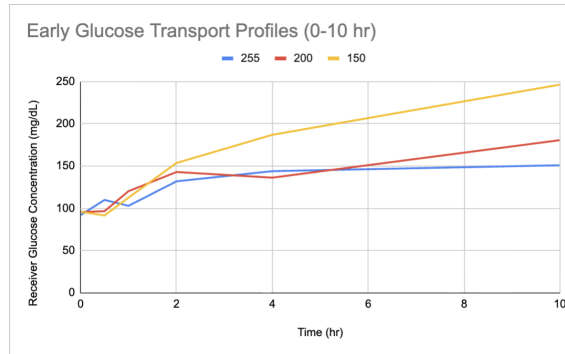


Figure 4.9: Early-time view of receiver glucose accumulation highlighting initial differences in transport kinetics between grayscale conditions.

All conditions demonstrated increasing receiver-phase glucose concentration over time, confirming successful diffusion through the hydrogel barrier. However, transport behavior varied substantially with grayscale condition. Lower-grayscale hydrogels exhibited faster increases in concentration and higher intermediate-time glucose levels than higher-grayscale samples.

At the 10-hour time point, mean receiver concentrations are shown in Table 4.2.

By 48 hours, the rank-order trend remained preserved, with the lowest grayscale condition exhibiting the greatest total glucose transport.

These results indicate that decreasing grayscale value increases effective permeability of the hydrogel barrier. This behavior is consistent with reduced crosslink density and

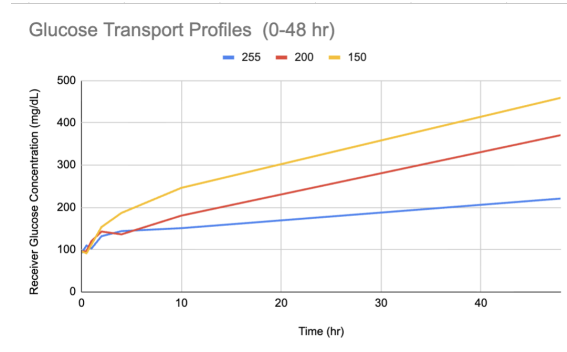


Figure 4.10: Mean receiver-phase glucose concentration over time following donor exposure. Lower grayscale hydrogels exhibited faster and greater glucose transport.

Table 4.3: Effective Diffusivity of Grayscale Hydrogels Normalized to 255

255	200	150
1.0	1.23	1.70

larger effective mesh size, which would permit more rapid solute diffusion through the polymer network.

Although replicate variability increased at later time points, particularly at 48 hours, consistent monotonic trends were observed across conditions. The persistence of these rank-order differences supports the conclusion that grayscale exposure reliably modulates hydrogel transport behavior.

4.3.3 Thickness-Corrected Effective Transport Behavior

Because lower grayscale hydrogels also exhibited reduced final thickness, part of the observed transport increase could arise from a shorter diffusion path length rather than changes in intrinsic permeability alone. To separate these effects, transport data were corrected using measured final hydrogel thicknesses.

A relative transport index proportional to effective diffusivity was estimated using the 10-hour concentration rise and final hydrogel thickness. Results are shown in Figure 4.10.

When normalized to the 255 condition:

Thus, even after accounting for barrier thickness, lower grayscale hydrogels retained substantially greater effective transport.

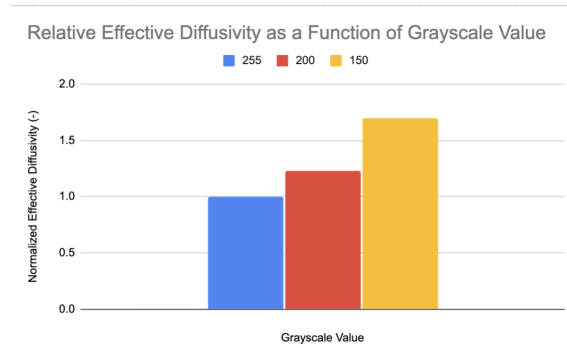


Figure 4.11: Normalized transport index proportional to effective diffusivity after correction for final hydrogel thickness. Lower grayscale conditions retained greater apparent intrinsic transport.

This finding indicates that grayscale exposure alters hydrogel transport through two coupled mechanisms:

Geometric effects: thinner barriers reduce diffusion distance
 Material effects: lower crosslinking density increases intrinsic permeability

Accordingly, grayscale-controlled photopolymerization provides a practical means of tuning hydrogel transport properties without altering device geometry or polymer formulation.

4.3.4 Implications for Capillary Barrier Mimicry

The ability to modulate glucose transport through grayscale exposure is directly relevant to the objective of replicating physiological transport barriers such as capillary basement membranes. In vivo, nutrient delivery is regulated not only by flow conditions but also by selective structural barriers that delay and attenuate solute transport.

The present results demonstrate that hydrogel barriers can be tuned across a range of transport behaviors through fabrication control alone. This enables selection of grayscale conditions that approximate desired physiological permeability ranges or generate intentionally altered transport states for disease modeling.

As a result, grayscale-defined hydrogels provide a flexible platform for constructing transport-controlled microenvironments within microfluidic systems.

Chapter 5

Analysis and Discussion

5.1 Transport-Limited Microenvironments vs Conventional Models

Traditional in vitro systems typically assume that cells are exposed to spatially uniform and temporally instantaneous concentrations of nutrients and signaling molecules. While this simplifies experimental design, it neglects the role of transport processes that govern solute delivery in vivo. In physiological environments, molecules such as glucose are delivered through vascular networks and must diffuse across capillary barriers and interstitial space before reaching target cells. This introduces spatial gradients and temporal delays that are not captured in conventional culture systems. The microfluidic platform developed in this work addresses this limitation by introducing a transport barrier between the flow channel and the cell chamber. By confining convective flow to a vascular channel and requiring solutes to diffuse through a hydrogel barrier, the system separates delivery from exposure. As a result, cellular environments are shaped by transport dynamics rather than imposed concentrations alone. This distinction is particularly relevant in the context of metabolic diseases such as diabetes, where fluctuations in glucose concentration occur within a transport-mediated environment. The present system provides a framework for studying how such dynamics influence cellular response under more physiologically

relevant conditions.

5.2 Decoupling of Flow and Diffusion

A key design feature of the platform is the decoupling of convective and diffusive transport. Flow within the vascular channel is driven by hydrostatic pressure, producing steady, laminar conditions that establish a controlled boundary concentration at the hydrogel interface. Because the hydrogel prevents direct fluid flow into the cell chamber, solute transport downstream is governed exclusively by diffusion. This separation enables independent control of two critical variables: flow conditions, which determine the upstream concentration profile, and hydrogel transport properties, which regulate solute propagation into the cell region. By isolating these mechanisms, the system minimizes confounding effects that can arise in microfluidic systems where flow directly influences cellular exposure. This design allows observed differences in downstream conditions to be attributed to material and transport properties rather than hydrodynamic variability.

5.3 Hydrogel as a Tunable Transport Interface

The hydrogel barrier functions as the primary regulator of transport within the system. Unlike traditional membranes with fixed properties, the hydrogel can be engineered to exhibit tunable diffusivity through control of crosslinking density. In this work, grayscale-controlled photopolymerization is used to define discrete hydrogel conditions corresponding to different levels of crosslinking. Variation in crosslinking density alters the effective mesh size of the polymer network, which in turn affects solute diffusivity. As a result, the hydrogel enables controlled modulation of permeability without altering device geometry. This approach allows transport to be treated as an experimental variable rather than a fixed parameter. Additionally, the use of spatially defined grayscale exposure introduces the potential for structured heterogeneity within the hydrogel. While the present study focuses on discrete conditions, this methodology can be extended to generate spatially

varying transport properties that more closely resemble discontinuous biological interfaces such as sinusoidal capillaries.

5.4 Hydrogel Swelling and Equilibration Behavior

Prior to diffusion experiments, hydrogels were equilibrated in media to ensure stable geometry and hydration state. Measurements of thickness and mass over time indicate an initial transient period followed by stabilization, consistent with swelling-driven equilibration. Thickness measurements were more consistent than mass measurements, which exhibited variability likely associated with surface liquid retention and handling effects. As a result, thickness was used as the primary parameter for defining the effective transport distance in subsequent analysis. The observed stabilization of thickness over time suggests that hydrogels reach approximate equilibrium within a finite timescale, after which changes are within measurement variability. Establishing this equilibrium state is critical, as diffusion behavior depends strongly on the barrier’s characteristic length scale.

5.5 Implications for Transport and Cellular Exposure

By introducing a tunable diffusion barrier, the system creates a microenvironment in which cellular exposure is governed by both time and position. Rather than experiencing a uniform concentration, cells are subjected to gradients that evolve as solutes diffuse through the hydrogel. This has important implications for experimental design and interpretation. In transport-limited systems, increasing the concentration of a solute in the upstream channel does not immediately translate to increased exposure at the cellular level. Instead, exposure depends on the interplay between delivery and diffusion, which may result in delayed or attenuated responses. Such effects are particularly relevant for evaluating engineered biological systems, including stem cell spheroids and biomaterial-based therapies, where local microenvironmental conditions play a critical role in function and viability. The present platform provides a means to systematically investigate these

effects under controlled transport conditions.

5.6 Experimental Limitations

Several limitations of the current system should be noted. First, hydrostatic pressure-driven flow, while simple and robust, may exhibit slight temporal variation as reservoir levels change. However, these effects are minimized through the use of sufficiently large reservoir volumes and consistent setup conditions. Second, while the hydrogel is treated as locally isotropic, spatial variations in crosslinking introduce heterogeneity that is not fully captured in simplified transport models. More detailed characterization of spatially varying diffusivity would improve predictive accuracy. Third, the current study isolates diffusion-driven transport and does not account for cellular consumption of solutes. In biological systems, uptake and metabolism introduce additional gradients that couple transport with reaction kinetics. Incorporating such effects represents an important direction for future work. Finally, while preliminary imaging demonstrates compatibility of the hydrogel environment with cells, comprehensive biological validation within the full transport-controlled system remains to be completed.

Chapter 6

Conclusion and Future Work

6.1 Summary of Work and Key Findings

This work presented the design and development of a microfluidic platform to study transport-limited solute delivery in a physiologically inspired environment. The system was constructed to decouple convective transport within a vascular channel from diffusive transport across a hydrogel barrier, enabling controlled investigation of how solutes propagate from a flowing medium into a downstream region. Hydrodynamic characterization confirmed that the flow within the vascular channel is laminar and stable, with predictable velocity profiles and pressure gradients. Hydrostatic pressure-driven flow was shown to provide a simple and effective method to maintain perfusion without the need for external pumping systems. Experimental validation demonstrated an average hydrostatic flow rate of approximately 10 $\mu\text{L}/\text{min}$, corresponding to a mean channel velocity of 0.40 mm/s under standard operating conditions. Hydrogel characterization established the importance of equilibration prior to transport experiments. Thickness measurements indicated that hydrogels reached an approximately stable hydrated state within the first 1–2 hours, after which subsequent changes were comparatively small and remained within normal measurement variability. Final equilibrated thickness depended strongly on grayscale exposure condition. Mean hydrated thicknesses were 0.768 mm (255), 0.656 mm (200), and 0.511 mm (150). These values were used as the characteristic diffusion

path lengths for subsequent transport analysis. Diffusion studies across hydrogels of varying grayscale-defined crosslinking density revealed clear differences in transport behavior. Receiver-phase glucose concentration increased over time for all conditions, confirming successful solute transport across the barrier. At 10 hours, mean receiver concentrations were 151 mg/dL (255), 181 mg/dL (200), and 246 mg/dL (150). By 48 hours, concentrations further increased to approximately 220 mg/dL (255), 370 mg/dL (200), and 460 mg/dL (150), indicating progressively greater cumulative transport at lower grayscale values. These results demonstrate that higher grayscale exposure corresponded to lower effective permeability, while reduced grayscale exposure increased solute transport. After correcting for thickness differences, the normalized relative effective diffusivity values were 1.00 (255), 1.23 (200), and 1.70 (150). This indicates that transport differences were not caused solely by barrier thickness, but also by grayscale-dependent changes in intrinsic hydrogel permeability. Collectively, these findings demonstrate that grayscale-controlled photopolymerization provides an effective method for tuning transport properties within microfluidic systems. Collectively, these findings demonstrate that the platform is capable of generating controlled, transport-limited microenvironments in which solute delivery is governed by both flow and diffusion. Looking at the objectives laid out by this project, most have been met. First, this project set out to fabricate a leak-resistant chip, which was completed by testing the chip qualitatively for leaks during perfusion. Next, passive perfusion was validated to produce stable and consistent flow by flow validation. Hydrogels were shown to have tunable transport properties by altering grayscale values and measuring effective diffusion. Finally, the goal of producing a platform suitable for future cell-based studies was partially met. The platform requires future diffusion testing with a cell monolayer and tissue cell experiments, which were outside the scope of the present work.

6.2 Significance of Transport-Controlled Microenvironments

A central contribution of this work is the explicit incorporation of a transport barrier into a microfluidic system. In contrast to conventional *in vitro* models, which assume uniform exposure, the present platform enforces a separation between solute delivery and cellular exposure. This results in concentration profiles that evolve over time and space, more closely reflecting physiological conditions. *In vivo*, solutes such as glucose are delivered through vascular networks and must traverse both endothelial layers and underlying basement membranes before reaching target cells. These processes introduce delays, gradients, and attenuation that are not captured in traditional culture systems. By modeling the basement membrane as a tunable hydrogel barrier, this work provides a first step toward incorporating these transport phenomena into *in vitro* platforms. The ability to control transport independently of flow enables systematic investigation of how microenvironmental conditions influence cellular behavior. This is particularly important in the context of metabolic diseases, where fluctuations in solute concentration occur within a transport-mediated framework. The present system, therefore, offers a more physiologically relevant approach to studying disease mechanisms and therapeutic response.

6.3 Future Work

6.3.1 Diffusion Characterization with Endothelial Layers

While the current system models diffusion across a capillary-like basement membrane using a hydrogel barrier, it does not yet incorporate the endothelial layer that forms the primary interface between blood flow and tissue. *In vivo*, solute transport across capillaries occurs through multiple pathways, including diffusion across endothelial cells, transport through tight junctions, and transendothelial channels. Future work will extend the platform by incorporating an endothelial monolayer at the interface between the vascular channel and the hydrogel. This addition will enable investigation of how cellular barriers modulate transport in conjunction with the underlying hydrogel. In particular, the com-

bined effects of endothelial permeability and hydrogel diffusivity can be studied to better approximate physiological transport across sinusoidal capillaries. Quantitative comparison between hydrogel-only and endothelial-integrated systems will provide insight into the relative contributions of basement membrane and cellular transport pathways. This represents a critical step toward developing a more complete and biologically representative capillary model.

6.3.2 Influence of Residence Time on Transport Behavior

The present system introduces a serpentine vascular channel designed to increase residence time and enhance interaction between the flowing medium and the diffusion interface. However, the relationship between residence time and downstream concentration profiles has not yet been fully characterized. Future studies will investigate how variations in flow rate and channel geometry affect residence time and, consequently, solute delivery to the hydrogel interface. By systematically varying hydrostatic pressure and channel length, it will be possible to determine how upstream transport conditions influence diffusion-driven exposure. Understanding this relationship is important for distinguishing between systems in which transport is limited by diffusion through the barrier and those in which upstream delivery plays a dominant role. This analysis will further clarify the interplay between convection and diffusion in shaping microenvironmental conditions.

6.3.3 Role of Transient Concentration Spikes vs Diffusion-Limited Exposure

In conventional *in vitro* experiments, cells are often exposed to rapid changes in solute concentration, resulting in transient spikes that may not accurately reflect *in vivo* conditions. In contrast, transport-limited systems introduce delays and gradients that may attenuate or redistribute these changes. Future work will examine whether cellular responses are driven primarily by peak concentration exposure or by the time-dependent diffusion profile. By comparing systems with direct exposure to those with diffusion-mediated delivery,

it will be possible to assess the relative importance of transient spikes versus sustained gradients. This distinction is particularly relevant for studying metabolic signaling and therapeutic delivery, where the timing and magnitude of exposure may significantly influence cellular outcomes.

6.3.4 Evaluation of Stem Cell Spheroids in Transport-Controlled Environments

The next phase of this work will involve integration of stem cell spheroids into the microfluidic platform to evaluate their behavior under transport-limited conditions. These spheroids, which are designed to retain stemness and support tissue regeneration through secretome modulation, represent a promising therapeutic approach for conditions such as diabetic cartilage degeneration. By placing spheroids within the diffusion-controlled chamber, the platform can be used to assess how transport dynamics influence cellular viability, signaling, and therapeutic efficacy. In particular, the system enables investigation of how limited nutrient delivery and gradient formation affect spheroid function. This application represents a shift from validating the platform itself to using it as a tool for evaluating emerging therapies under physiologically relevant conditions.

6.4 Final Conclusions

This work demonstrates the feasibility of engineering a microfluidic system in which solute transport is explicitly controlled through the integration of convective flow and diffusive barriers. By combining hydrostatic flow, tunable hydrogel interfaces, and spatially defined transport conditions, the platform establishes a framework for studying how microenvironmental dynamics influence biological systems. A key insight of this work is that transport barriers play a fundamental role in shaping cellular exposure. In physiological systems, cells do not experience uniform concentrations, but rather gradients governed by the interplay of flow, diffusion, and structural interfaces. Incorporating these effects into in vitro models is essential for improving their relevance and predictive capability. The

platform developed here provides a step toward this goal by enabling controlled, tunable transport conditions that more closely resemble those found *in vivo*. While further work is required to incorporate additional biological complexity, including endothelial layers and cellular consumption, the system establishes a foundation for future studies in drug testing, tissue engineering, and disease modeling. Ultimately, the integration of transport phenomena into microfluidic systems represents a critical advancement in the development of physiologically realistic models. By shifting the focus from static exposure to dynamic, transport-mediated environments, this work contributes to a more accurate understanding of how biological systems respond to their surroundings.

Appendix A: Computational Modeling Methods and Repository

Full CFD simulation files, geometry files, and supporting analysis materials associated with this thesis are available in the public project repository:

<https://github.com/jlevin8/microfluidicchipcapstone2026>

Repository contents include simulation geometries, boundary condition definitions, mesh files, and exported post-processing results sufficient to reproduce the computational analyses presented in Chapter 4.

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